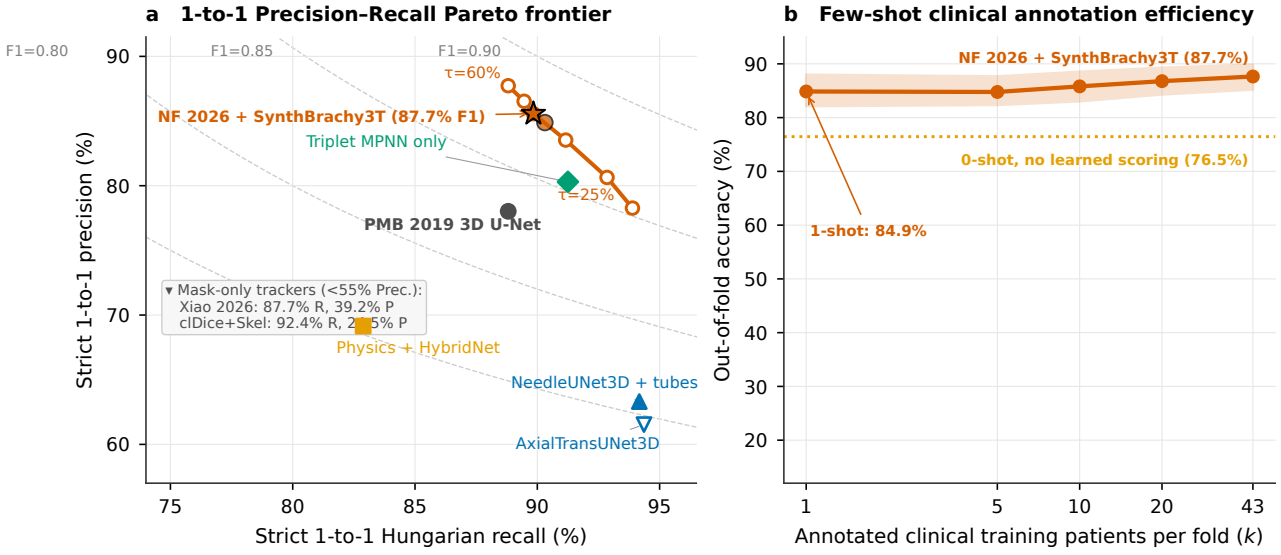
a 1-to-1 Precision–Recall Pareto frontier
F1=0.80
F1=0.85
F1=0.90
τ=60%
NF 2026 + SynthBrachy3T (87.7% F1)
Triplet MPNN only
τ=25%
PMB 2019 3D U-Net
▾ Mask-only trackers (<55% Prec.):
Xiao 2026: 87.7% R, 39.2% P
clDice+Skel: 92.4% R, 2 5% P
Physics + HybridNet
NeedleUNet3D + tubes
AxialTransUNet3D
Strict 1-to-1 precision (%)
Strict 1-to-1 Hungarian recall (%)
90
80
70
60
75
80
85
90
95
b Few-shot clinical annotation efficiency
NF 2026 + SynthBrachy3T (87.7%)
0-shot, no learned scoring (76.5%)
1-shot: 84.9%
Out-of-fold accuracy (%)
Annotated clinical training patients per fold (k)
90
80
70
60
50
40
30
20
1
5
10
20
43